

Assignment Instructions:

- Work **individually**; plagiarism will result in zero marks.
- Use **clear diagrams** and label all components properly.
- Attempt all questions.
- Write concise yet complete answers.

Topics: (Intro to OS, Structure & Process Management).

[Mapped with CLO-01]

Part A: Introduction to Operating Systems.

1. Scenario – Choosing the Right OS

You are setting up a new computer lab with 30 systems for programming and design classes.

- Which type of operating system (Batch, Time-Sharing, Real-Time, Distributed) would you select and why?
- List **three essential OS services** required in this environment.

2. Diagram Question – OS as a Layered System

Draw and explain a labeled diagram showing the **relationship between hardware, operating system, and application software**.

- Briefly describe the role of each layer in 2–3 lines.

Part B: Operating System Structures.

3. Scenario – Server Crash

A university server running a **Monolithic OS** crashes frequently due to faulty drivers.

- Compare **Monolithic**, **Modular**, and **Microkernel** structures.
- Which structure would you recommend to reduce system crashes, and why?

4. Diagram Question – OS Structures

Draw a **labeled diagram** showing the **Layered OS architecture**.

- Identify the main functions at each layer (e.g., hardware control, system calls, user interface).

5. Scenario – Smartphone OS

A smartphone company is designing a new OS that supports multimedia, sensors, and regular updates.

- Which OS structure (Layered, Modular, or Hybrid) would be most efficient? Justify with two points.

Part C: Process Management & Interrupts

6. Scenario – Video Streaming Lag

A video streaming app slows down when multiple downloads occur in the background.

- Explain how **CPU scheduling** and **process states** are involved.
- Suggest how **priority scheduling** could improve user experience.

7. Scenario – Banking Transactions

Multiple users are performing online banking transactions simultaneously.

- Explain the importance of **process synchronization** in this case.
- What problems can occur if synchronization fails? Give an example.

8. Interrupts – Concept & Example

- Define **interrupts** and explain their role in OS process management.
- Give **one hardware** and **one software interrupt** example from real life (e.g., keyboard input, divide-by-zero error).

9. Diagram Question – Process State Diagram

Draw and label a **process state transition diagram** showing:

- New → Ready → Running → Waiting → Terminated states
- Explain each transition briefly.

10. Scenario – Real-Time System

A hospital emergency system must handle life-critical alerts instantly.

- Which type of OS and **scheduling algorithm** would best suit this?
- Justify your answer.

End of Assignment

Submit on time and ensure your answers demonstrate conceptual understanding and real-world application.